



Week 02 Task-05

Networking Tools and Subnetting Basics

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Reference Number: TBS-INT-2026-445

Submission Date: 27|8|2026

Submitted To: TechBiz Security

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1) Executive Summary

Networking fundamentals play a key role in daily life of a cybersecurity professionals, as understanding how systems communicate is essential for performing security assessments and troubleshooting network-related issues. This task focused on practicing essential networking commands such as ping, traceroute, dig, and nslookup to understand practical networking concepts. Subnetting is also practiced to understand the importance of subnetting how segmentation is done in computer networks for a security purpose what does a CIDR (**C**lasseless **i**nter **d**omain **r**outing) notation represents in subnetting.

2) Scope

The scope of this task is to learn about networking protocols such as network connectivity how http and dns works how subnetting is used for better network utilization and security purpose.

3) Tools used

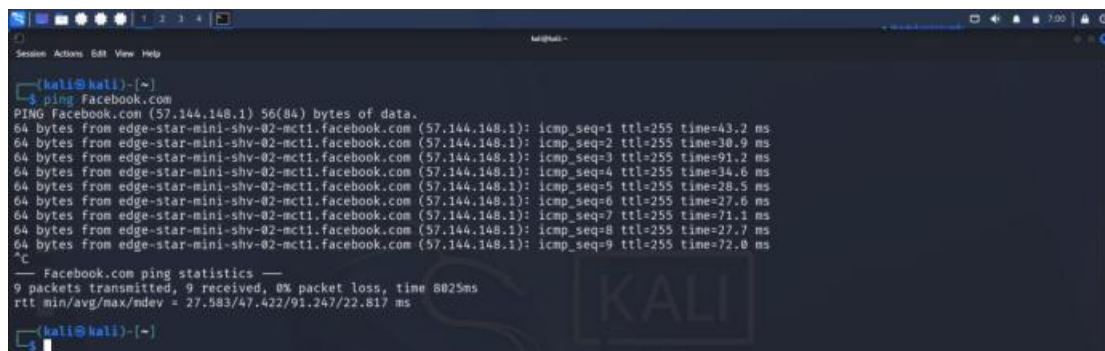
- Kali Linux
- TryHackMe room DNS in Detail
- TryHackMe room HTTP in Detail

4) Ping Command Practical Execution

4.1) Purpose of ping command

The Purpose of ping command is use to check the network connectivity. It is also use to determine if target system is alive or not.

4.2) Ouput of ping command



```
(kali@kali)~$ ping Facebook.com
PING Facebook.com (57.144.148.1) 56(84) bytes of data:
64 bytes from edge-star-mini-shv-02-mct1.facebook.com (57.144.148.1): icmp_seq=1 ttl=255 time=43.2 ms
64 bytes from edge-star-mini-shv-02-mct1.facebook.com (57.144.148.1): icmp_seq=2 ttl=255 time=30.9 ms
64 bytes from edge-star-mini-shv-02-mct1.facebook.com (57.144.148.1): icmp_seq=3 ttl=255 time=91.2 ms
64 bytes from edge-star-mini-shv-02-mct1.facebook.com (57.144.148.1): icmp_seq=4 ttl=255 time=34.6 ms
64 bytes from edge-star-mini-shv-02-mct1.facebook.com (57.144.148.1): icmp_seq=5 ttl=255 time=28.5 ms
64 bytes from edge-star-mini-shv-02-mct1.facebook.com (57.144.148.1): icmp_seq=6 ttl=255 time=27.6 ms
64 bytes from edge-star-mini-shv-02-mct1.facebook.com (57.144.148.1): icmp_seq=7 ttl=255 time=71.1 ms
64 bytes from edge-star-mini-shv-02-mct1.facebook.com (57.144.148.1): icmp_seq=8 ttl=255 time=27.7 ms
64 bytes from edge-star-mini-shv-02-mct1.facebook.com (57.144.148.1): icmp_seq=9 ttl=255 time=72.0 ms

--- Facebook.com ping statistics ---
 9 packets transmitted, 9 received, 0% packet loss, time 6025ms
rtt min/avg/max/mdev = 27.583/47.422/91.247/22.817 ms
```

The ping command successfully confirm that network connectivity is working properly shown in above Image because there is no packet loss all packets that send received also. Ping command uses the icmp (**I**nternet **C**ontrol **M**essage **P**rotocol)

echo requests packets to the destination server and wait for reply. IP address after resolving dns name Facebook the IP is (57.144.148.1)

Output Analysis

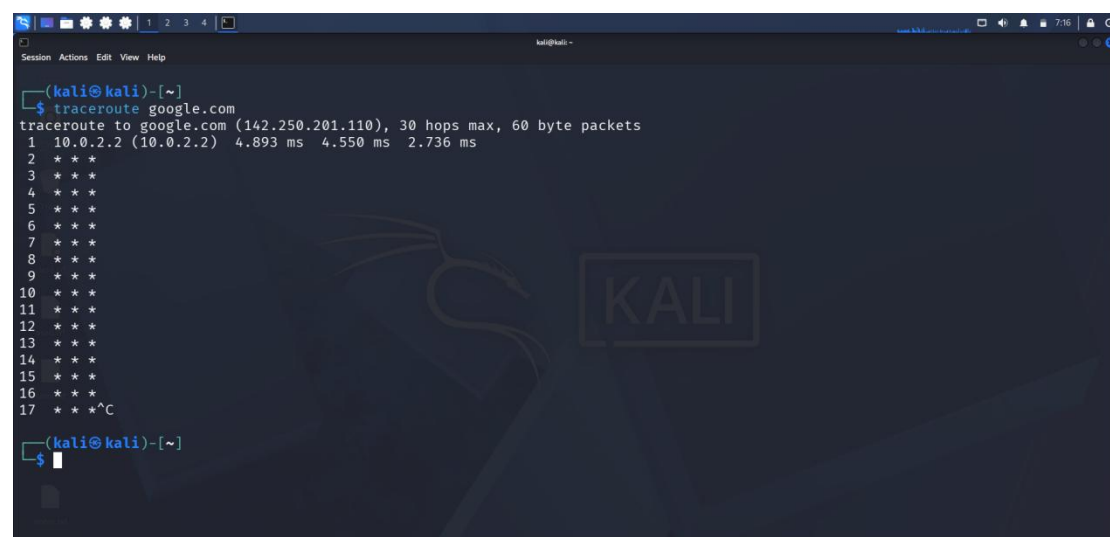
- **Size of ICMP response** = 64 bytes
- **time to live value (ttl)** = 255
- **icmp_seq** = each number of icmp request is visible in a sequence
- **Time** = 8025ms

5) Traceroute Command Practical Execution

5.1) Purpose of Traceroute command

Traceroute is a network diagnostic command used to identify the path taken by data packets from source to destination server. It shows intermediate router hops through which packet travels and measure the latency for each hop.

5.2) Output of Traceroute command



```
(kali@kali)-[~]
$ traceroute google.com
traceroute to google.com (142.250.201.110), 30 hops max, 60 byte packets
 1  10.0.2.2 (10.0.2.2)  4.893 ms  4.550 ms  2.736 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * * ^C

(kali@kali)-[~]
$
```

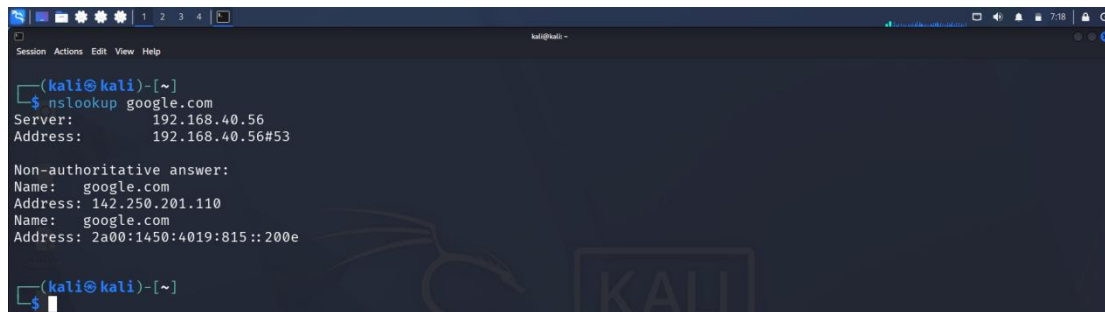
Traceroute command successfully resolved google domain name into ip address. **142.250.201.110**. In the above figure it is seen that first network hop is successfully received. Router didn't not response to other traceroute request. All other traceroute requests gets blocked because many network devices block ICMP /UDP traceroute for security reasons and don't reveal actual information.

6) DNS Lookup using nslookup Practical Execution

6.1) Purpose of dns lookup using nslookup

The main Purpose of dns lookup using nslookup (Name Server Lookup) is to query DNS servers to retrieve mapping information between domain name and ip addresses.

6.2) Output of dns lookup using nslookup

A screenshot of a Kali Linux terminal window. The prompt is (kali@kali)-[~]. The user has entered the command nslookup google.com. The output shows the server used (192.168.40.56) and the IP address (192.168.40.56#53). It then displays a non-authoritative answer for google.com, showing the IPv4 address 142.250.201.110 and the IPv6 address 2a00:1450:4019:815::200e.

```
(kali@kali)-[~]  
$ nslookup google.com  
Server:      192.168.40.56  
Address:     192.168.40.56#53  
  
Non-authoritative answer:  
Name:   google.com  
Address: 142.250.201.110  
Name:   google.com  
Address: 2a00:1450:4019:815::200e  
  
(kali@kali)-[~]  
$
```

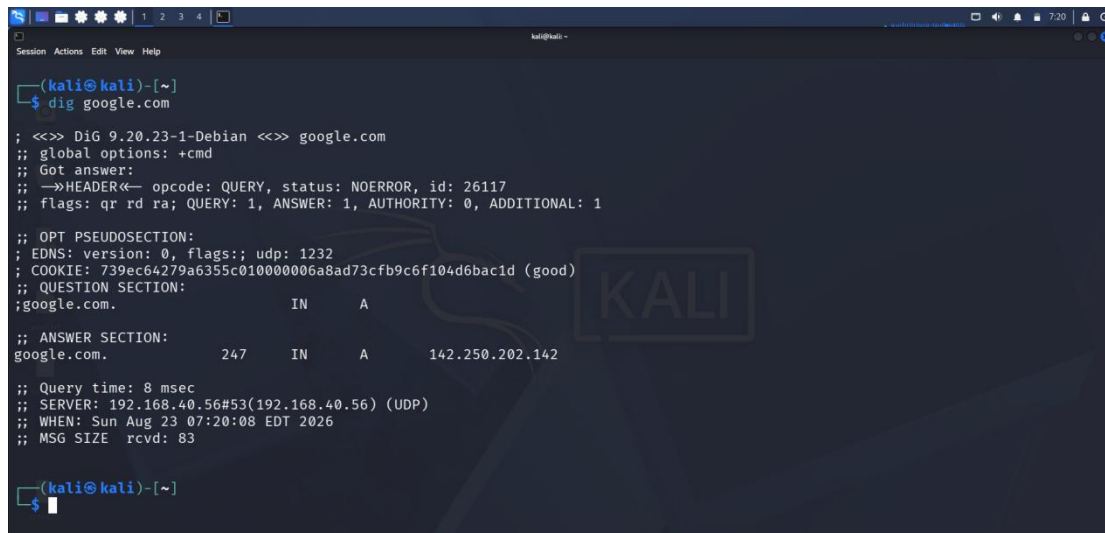
In the above Figure it is clearly shown that name google.com is successfully resolved into ip address. Both ipv4 (**142.250.201.110**) and ipv6 (**2a00:1450:4019:815::200e**) address of google server are available. **192.168.40.56** this is the local dns server answering my request.

7) DNS Lookup using dig Practical Execution

7.1) Purpose of dns lookup using dig

dig (Domain information gropper) is a tool use for dns lookup . It provides detailed raw network information about domain records. As compare to nslookup dig is the modern tool used by administrators and security professionals, It is useful in reconnaissance phase of ethical hacking by discovering hidden subdomains, mailservers and zone transfer vulnerabilities of organizations

7.2) Output of dns lookup using dig



```
(kali@kali)-[~]
$ dig google.com

;<<>> DiG 9.20.23-1-Debian <<>> google.com
;; global options: +cmd
;; Got answer:
-->HEADER<-- opcode: QUERY, status: NOERROR, id: 26117
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:: udp: 1232
; COOKIE: 739ec64279a6355c010000006a8ad73cfb9c6f104d6bac1d (good)
;; QUESTION SECTION:
;google.com.                IN      A
;; ANSWER SECTION:
google.com.                 247     IN      A      142.250.202.142

;; Query time: 8 msec
;; SERVER: 192.168.40.56#53(192.168.40.56) (UDP)
;; WHEN: Sun Aug 23 07:20:08 EDT 2026
;; MSG SIZE rcvd: 83

(kali@kali)-[~]
$
```

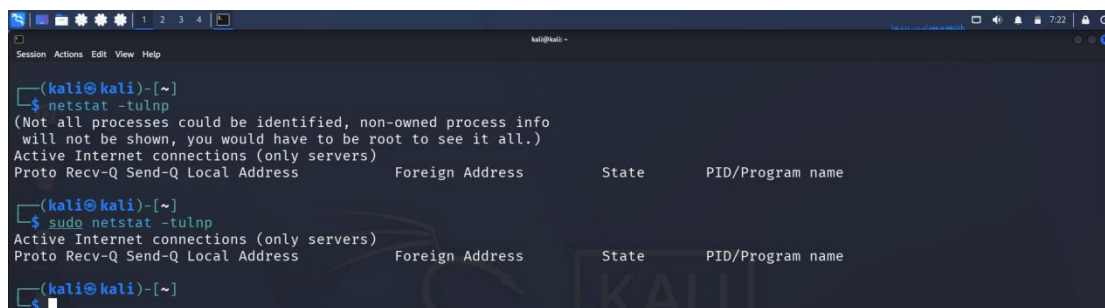
In the above picture it is clearly seen that domain name google is resolved into ip address of google using dig tool. The IP address of google server is 142.250.202.142. The header section indicates the status no error that means target domain exist and successfully resolved into IP address. The total packet size was 83 bytes. Query time 8msec.

8) Netstat Command Practical Execution

8.1) Purpose of netstat command

The netstat command (**Network statics**) is used to display active network connections, listening ports, and services running on the network. This command is highly useful in network monitoring for administrative purpose and identifying available services during security assessment.

8.2) Output of netstat command



```
(kali@kali)-[~]
$ netstat -tulnp
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
(kali@kali)-[~]
$ sudo netstat -tulnp
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
(kali@kali)-[~]
$
```

It is clearly shown in above Image that **netstat -tulnp** command is executed with root and without root privileges. It is found that there is no active listening ports and services were detected.

tuInp breakdown

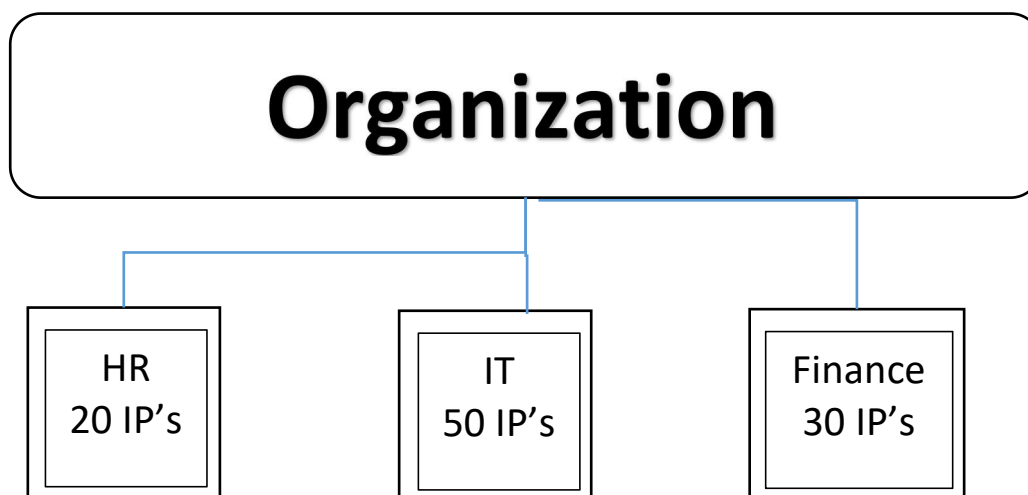
- t = tcp
- u = udp
- l = show listening ports
- n = show numerical addresses instead of resolving names.
- p = Display process/program using port

9) Subnetting Practical

Subnetting is the process of dividing a large ip network into smaller logical networks called subnets. Subnetting is performed due to following reasons mentioned below.

1. **Efficient IPV4 utilization:-** With the help of subnetting we can only allocate the required IP address to each department in this way we can avoid the wastage of IPv4 address.

Example



So now we have a IP network that has 254 usable IP's (**IPv4**) excluding 1 for broadcast and 1 for network id (network address).

Required IP's are $20+50+30 = 100$ IP's that means **154 IP's** are waste in 1 network and if this keeps happening in every **IP network** then the world will face shortage of ipv4 addresses.

To solve this issue we create a separate subnet for each department and in this way only required IP's will be used there will be no IP's address wastage. That's why subnetting is the efficient ipv4 utilization.

2. **Improved Security:** Now lets consider if cyber attack happens on a subnet **HR** then all other subnets are isolated means that they do not face attack consequences immediately although its on same network but hacker needs to spread his attack on entire network.

3. **Easier trouble shooting:** if there is an issue in a network because of subnetting network engineers will quickly recognize the main problem area not the entire network will be down. They will quickly identify and solve the issue.

Practice Example of IP network

192.168.1.0/24

Octet Structure of Class c IP network

192 . 168 . 1 . 0/24 — CIDR
N N N H

Look at the CIDR (**Classless Inter Domain Range**) its 24. this means that we have 24 **network Bits** and 8 **Host Bits**.

To calculate host bits the formula is pretty simple

Host bits = Total IP address bits - network bits

Host bits = 32 - 24

Host bits = 8

Total IP address are $2^8 = 256$

To calculate total usable IP address we can use the formula

Total Usable IP address = $2^{\text{Host bits}} - 2$

Total Usable IP address = $2^8 - 2$

Total Usable IP address = $256 - 2$

Total Usable IP address = 254

The reason of subtracting 2 from 256 is one is Network ID and one is Broadcast ID.

Hence this is a class c IP address therefore its default subnet Mask is **255.255.255.0**

Result

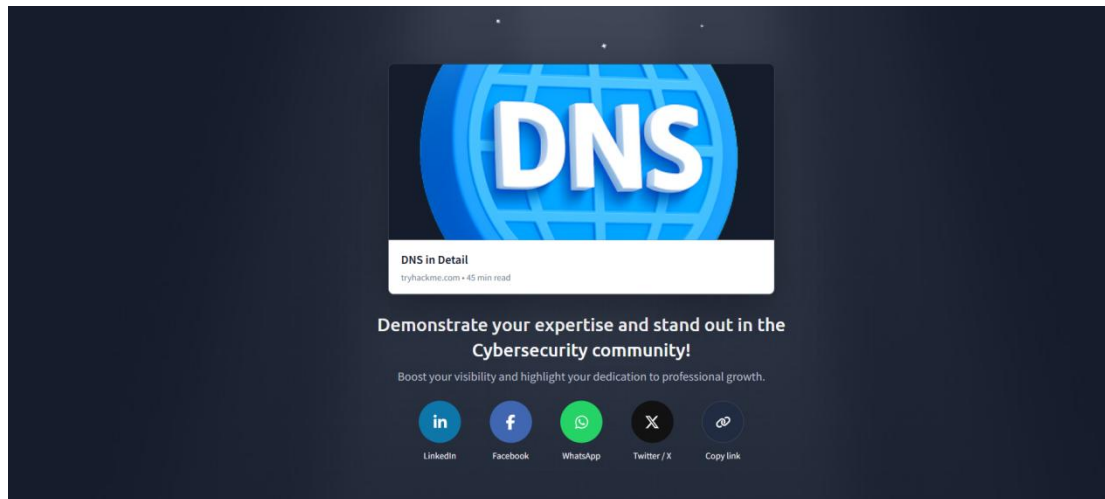
IP network = 192.168.1.0

Subnet Mask = 255.255.255.0

Total IP address = 256

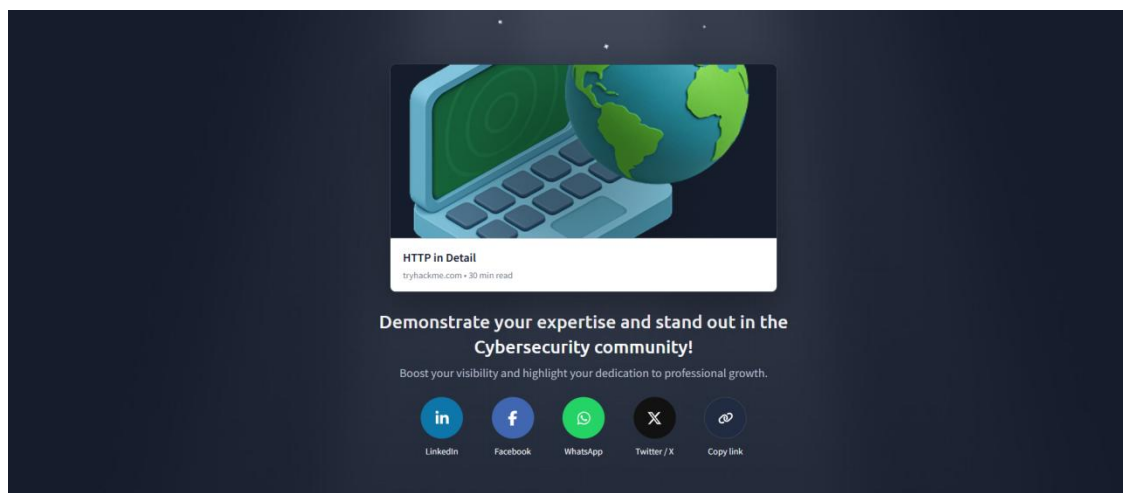
Usable IP address = 254
First Usable Host = 192.168.1.1
Last Usable Host = 192.168.1.254
Network ID = 192.168.1.0
Broadcast Address = 192.168.1.255

10) TryHackMe Room DNS in Detail



In the TryHackMe room DNS in Detail I learned about the complete working mechanism that how dns works. How domain name gets resolved into IP address. DNS record types such as A Record AAAA Record CNAME Record MX Record TXT Record. In this room I also learned about domain heirarchy like TLD (Top level Domain) Second Level Domain and Subdomain.

11) TryHackMe Room HTTP in Detail



In the TryHackMe room HTTP in Detail. I learned complete details about the http protocol how it works difference between http and https.I learned how request is

processed by web server and clients get the response back in case of any query. I also discovered the port number that http uses for the different conditions example 403 is not authorized. I discovered about http headers such as Content length content type Host name Agent. Different request for Data Handling like GET, POST, PUT, DELETE. GET is used to retrieve data from web server post is used to create data in web server delete is used to delete data from web server put is used to update data. I also learned the importance of cookies in web servers.

12) Youtube video Subnetting explained for beginners

Consider we have a big single network having IP Address 200.1.2.0. We want to do subnetting and divide this network into 4 subnets.

1st Subnet

- IP Address of the subnet / Subnet id = 200.1.2.0
- Direct Broadcast Address = 200.1.2.01111111 = 200.1.2.63
- Total number of IP Addresses = $2^6 = 64$
- Range of IP Addresses = [200.1.2.0, 200.1.2.63]
- Total number of hosts that can be configured = $64 - 2 = 62$
- Range of Allocated IP Addresses = [200.1.2.1, 200.1.2.62]

2nd Subnet

- IP Address of the subnet / Subnet id = 200.1.2.64
- Direct Broadcast Address = 200.1.2.01111111 = 200.1.2.127
- Total number of IP Addresses = $2^6 = 64$
- Range of IP Addresses = [200.1.2.64, 200.1.2.127]
- Total number of hosts that can be configured = $64 - 2 = 62$
- Range of Allocated IP Addresses = [200.1.2.65, 200.1.2.126]

3rd Subnet

- IP Address of the subnet / Subnet id = 200.1.2.128
- Direct Broadcast Address = 200.1.2.10111111 = 200.1.2.191
- Total number of IP Addresses = $2^6 = 64$
- Range of IP Addresses = [200.1.2.128, 200.1.2.191]
- Total number of hosts that can be configured = $64 - 2 = 62$
- Range of Allocated IP Addresses = [200.1.2.129, 200.1.2.190]

4th Subnet

- IP Address of the subnet / Subnet id = 200.1.2.192
- Direct Broadcast Address = 200.1.2.11111111 = 200.1.2.255
- Total number of IP Addresses = $2^6 = 64$
- Range of IP Addresses = [200.1.2.192, 200.1.2.255]
- Total number of hosts that can be configured = $64 - 2 = 62$
- Range of Allocated IP Addresses = [200.1.2.193, 200.1.2.254]

10 मिनट्स हो गए इसके बाद जब सारा जीरो रखोगे तो 64 कि जब

4.9 Complete Subnetting in One video

KnowledgeGATE by Sanchit Sir

13) Youtube video explaining ping traceroute nslookup

Ping command

YouTube video player showing a video titled "ping traceroute nslookup". The video content includes a diagram illustrating the Ping utility. A laptop is connected to a server labeled "216.109.117.108" via the "INTERNET". A text box states: "echo reply requests. And these replies will inform us". Below the diagram, the video title "PING and TRACERT (traceroute) networking commands" is displayed.

Command Prompt output for Ping:

```
Microsoft Windows [Version 3.1.2600]
(C) Copyright 1995-2000 Microsoft Corp.

C:\Documents and Settings\user> ping 216.109.117.108

Reply from 216.109.117.108: bytes=32 time=38ms TTL=47
Reply from 216.109.117.108: bytes=32 time=38ms TTL=47
Reply from 216.109.117.108: bytes=32 time=38ms TTL=47
Reply from 216.109.117.108: bytes=32 time=38ms TTL=47

Ping statistics for 216.109.117.108:
    Packets: Sent = 4, Received = 4, Lost = 0
```

Traceroute Command

YouTube video player showing a video titled "ping traceroute nslookup". The video content includes a diagram illustrating the Traceroute utility. A laptop is connected to a server labeled "66.176.124.54" via a series of routers. A text box states: "Such as the IP address and the time it took between each". Below the diagram, the video title "PING and TRACERT (traceroute) networking commands" is displayed.

Command Prompt output for Traceroute:

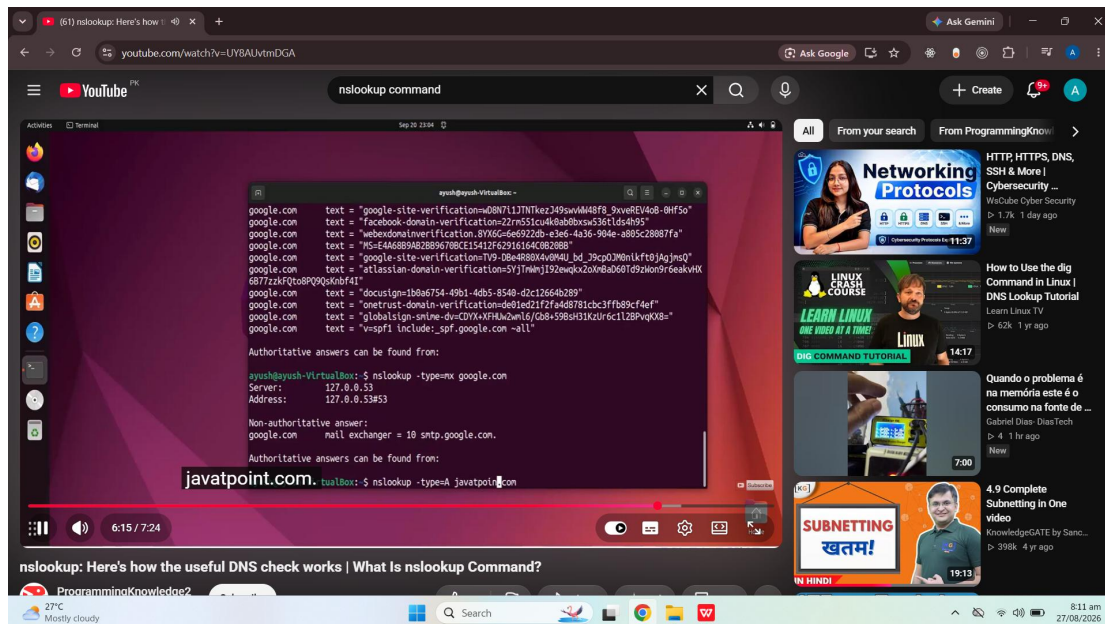
```
Microsoft Windows [Version 3.1.2600]
(C) Copyright 1995-2000 Microsoft Corp.

C:\Documents and Settings\user> traceroute 66.176.124.54

 1  8 ms  13 ms  12 ms  73.12.173.2
 2  6 ms  11 ms  12 ms  attga.ip.att.net [12.123.32.106]
 3  8 ms  10 ms  10 ms  cornll.ip.att.net [12.212.41.172]
 4  7 ms  10 ms  10 ms  drfmi92av.ip.net [66.12.2.212]
 5  7 ms  15 ms  13 ms  [66.176.124.54]

Trace complete.
```

nslookup Command



14) Conclusion

In conclusion I can say that this task helped me to understand the core networking concepts and commands use in security assessment. It also helped me to understand the importance of subnetting how subnetting works understanding the cidr notation in subnetting. Other than that through this practical task I improved my concepts regarding DNS and HTTP how dns works under the hood and how http and https is different, why https is preferred, different types of http methods, working of request and response. Over all this task enhanced my foundation for further cyber security tasks.